

Muhammad Nabeed Haider

Computer Science Undergraduate

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PROFESSIONAL SUMMARY

Computer Science undergraduate with hands-on experience developing Python applications, backend services, and AI-powered software. Built projects using Python, FastAPI, LangGraph, SQL, and modern development tools while following modular software design principles. Seeking a Junior Python Developer role to build reliable software and continue growing as a backend engineer.

EDUCATION

Bachelor of Science in Computer Science

2026

FAST National University of Computer and Emerging Sciences, Islamabad

- CGPA: 3.56 / 4.00
- Relevant Coursework: Generative AI, Machine Learning, MLOps, Natural Language Processing, Statistical Modelling, Cloud Computing, Information Security

EXPERIENCE

AI Lab Demonstrator

Spring 2026

FAST National University of Computer and Emerging Sciences

- Conducted weekly Artificial Intelligence laboratory sessions for approximately 50 undergraduate students.
- Provided guidance on Python programming, search algorithms, machine learning fundamentals, and debugging techniques.
- Assisted with grading programming assignments, evaluating lab performance, and supporting students during practical sessions.
- Collaborated with the course instructor to maintain a consistent and engaging laboratory environment.

PROJECTS

GymBuddy

2026

Python • FastAPI • Ollama • RAG

- Developed backend REST APIs for an AI-powered fitness assistant using FastAPI.
- Integrated locally hosted Ollama language models to provide context-aware fitness recommendations.
- Implemented a Retrieval-Augmented Generation (RAG) workflow to improve response relevance using external knowledge.
- Collaborated within a team to design, test, and demonstrate backend functionality through a lightweight web interface.

SCAR — Multimodal AI Assistant

2026

Python • FastAPI • Ollama • n8n • RAG • LoRA

- Developed a Jarvis-inspired multimodal AI assistant capable of conversational interaction, image generation, retrieval-augmented reasoning, and local system automation.
- Implemented a Retrieval-Augmented Generation (RAG) pipeline using n8n to retrieve and incorporate knowledge from Google Drive documents into AI responses.

- Fine-tuned Stable Diffusion 1.5 using LoRA to support customized image generation alongside local LLM-powered conversational capabilities.
- Integrated external services including weather, sports, and news APIs to provide real-time information within a unified assistant interface.
- Designed a modular architecture enabling future expansion through additional tools, APIs, and multi-modal capabilities.

Production MLOps Pipeline

2026

Python • FastAPI • Docker • Prometheus • Grafana

- Developed a production-oriented machine learning inference service using FastAPI and Docker.
- Integrated Prometheus metrics and Grafana dashboards to monitor application health and model-serving performance.
- Implemented CI workflows using GitHub Actions to automate testing and deployment tasks.
- Applied modular software architecture and containerization practices to improve deployment reproducibility and maintainability.

FakeLess — Audio Deepfake Prevention

2026

Python • PyTorch • SpeechBrain • HiFi-GAN

- Designed and developed an audio protection framework that reduces the effectiveness of neural voice cloning while preserving perceptual speech quality.
- Implemented a deep learning training pipeline using PyTorch with SpeechBrain speaker embeddings and HiFi-GAN-based audio reconstruction.
- Processed large-scale LibriSpeech datasets through automated preprocessing and feature extraction pipelines for model training.
- Evaluated system performance using objective speech-quality and speaker-similarity metrics including PESQ, STOI, and embedding similarity.
- Developed an end-to-end training pipeline for universal adversarial perturbation learning using LibriSpeech and speaker embedding optimization.

TECHNICAL SKILLS

Languages: Python, Java, SQL

Backend Development: FastAPI, REST APIs

Artificial Intelligence: Ollama, LangChain, LangGraph, Retrieval-Augmented Generation (RAG)

Databases: SQLite, ChromaDB

Tools: Git, GitHub, Docker, Linux

Concepts: Object-Oriented Programming, Data Structures and Algorithms, API Development, Modular Software Design

ACHIEVEMENTS

- AI Lab Demonstrator for the Artificial Intelligence course at FAST-NUCES.
- Developed multiple end-to-end AI and machine learning projects spanning Generative AI, MLOps, and backend systems.
- Hands-on experience with production-oriented machine learning workflows, CI/CD, Docker, monitoring, and LLM-based applications.